

DATA SET

DS1406

10. Marriage Rates Half of adults today are married compared with the all-time high of 72% in 1960.⁹ Do marriage rates vary by educational level? A sample of 300 adults in each of three educational categories provided the following information.

Educational Level	Married	
	Yes	No
Bachelors or higher	187	113
Some college	162	138
High school or less	149	151

498 402 900

Test to determine whether marriage rates differ significantly among adults in these educational levels using $\alpha = .01$. How would you summarize your results?

H_0 : marriage rate 與教育程度無關
 H_a : marriage rate 因教育程度不同而不同

期望值:

$$E_{yes} = \frac{300(498)}{900} = 166$$

$$E_{no} = \frac{300(402)}{900} = 134$$

$$\chi^2 = \sum \frac{(O-E)^2}{E} = 10.161$$

$$= \frac{(187-166)^2}{166} + \dots + \frac{(112-134)^2}{134} + \dots + \frac{(151-134)^2}{134}$$

$$df = (3-1)(2-1) = 2$$

$$\alpha = 0.01 \quad \chi^2_{0.01, 2} = 9.210 < 10.161$$

$\therefore$ reject H_0

有顯著差異

DATA SET

DS1407

12. How Big Is the Household? A local chamber of commerce surveyed 120 households in their city—40 in each of three types of residence (apartment, duplex, or single residence)—and recorded the number of family members in each of the households. The data are shown in the table.

Family Members	Type of Residence		
	Apartment	Duplex	Single Residence
1	8	20	1
2	16	8	9
3	10	10	14
4 or More	6	2	16
	40	40	40

Is there a significant difference in the family size distributions for the three types of residence? Test using $\alpha = .01$. If there are significant differences, describe the nature of these differences.

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

$$= \frac{(8-9.667)^2}{9.667} + \frac{(20-9.667)^2}{9.667} + \frac{(1-9.667)^2}{9.667} + \dots$$

$$E = \frac{29(40)}{120} = 9.667$$

$$\frac{30(40)}{120} = 10$$

$$\frac{34(40)}{120} = 11.33$$

$$\frac{24(40)}{120} = 8$$

H_0 : 三種住宅的家庭人數分布相同

H_a : 不同

$$\Rightarrow \chi^2 = 36.499$$

$$df = (4-1)(3-1) = 6$$

$$\chi^2_{0.01, 6} = 16.812 < 36.499$$

所以 reject H_0 , 有顯著差異

$$E = \frac{(\text{row total})(\text{column})}{\text{grand total}}$$

12. Dissolved Oxygen Content Dissolved oxygen content is a measure of the ability of a river, lake, or stream to support aquatic life, with high levels being better. A pollution-control inspector who suspected that a river community was releasing semitreated sewage into a river, randomly selected five specimens of river water at a location above the town and another five below. These are the dissolved oxygen readings (in parts per million):

Above Town: 4.8, 5.2, 5.0, 4.9, 5.1 $\Rightarrow 34$

Below Town: 5.0, 4.7, 4.9, 4.8, 4.9 $\Rightarrow 21$

- a. Use a one-tailed Wilcoxon rank sum test with $\alpha = .05$ to confirm or refute the theory.
b. Use a Student's t -test (with $\alpha = .05$) to analyze the data. Compare the conclusion reached in part a.

(a) H_0 : above, below 分布相同

H_a : below town 的 dissolved oxygen 较低

合併排序後, below town 的 rank sum

$$5 + 4.9 + 4.9 + 4.8 + 4.9$$

$$W_0 = 21 = 1 + 2 + 5 + 10 + 15$$

$$n_1 = n_2 = 5 \quad \alpha = 0.05$$

$$P(W \leq 21) = \frac{28}{C_{10}^5} = 0.111$$

≥ 0.05

$\therefore$ Non-reject H_0

沒有足夠 evidence 支持

below town 的 dissolved oxygen 含量較低

reject if $W \geq 36$
 $5.5, 0.05$

(b) Above town Avg = $\bar{X}_A = 5$ $S_A^2 = 0.025$

Below town Avg = $\bar{X}_B = 4.86$ $S_B^2 = 0.013$

$$s_p^2 = \frac{(5-1)(0.025) + (5-1)(0.013)}{5+5-2} = 0.019$$

$$t = \frac{\bar{X}_A - \bar{X}_B}{s_p \sqrt{\frac{1}{n_A} + \frac{1}{n_B}}} = \frac{5 - 4.86}{\sqrt{0.019} \sqrt{\frac{1}{5} + \frac{1}{5}}} = 1.606$$

$$< t_{0.05, 8} = 1.86$$

$\therefore$ fail to reject H_0

Wilcoxon rank-sum test, Student t -test same conclusion

not enough evidence at $\alpha = 0.05$ that dissolved oxygen is lower below town

DATA SET DS1502

13. Eye Movement In an investigation of the visual scanning behavior of deaf children, measurements of eye movement were taken on nine deaf and nine hearing children. The table gives the eye-movement rates and their ranks (in parentheses). Does it appear that the distributions of eye-movement rates for deaf children and hearing children differ?

Deaf Children	Hearing Children
2.75 (15)	.89 (1)
2.14 (11)	1.43 (7)
3.23 (18)	1.06 (4)
2.07 (10)	1.01 (3)
2.49 (14)	.94 (2)
2.18 (12)	1.79 (8)
3.16 (17)	1.12 (5,5)
2.93 (16)	2.01 (9)
2.20 (13)	1.12 (5,5)
Rank Sum	126
	45

$n_D = 9$

$n_H = 9$

$W_H = 45$

$W_D = 126$

$N = 18$

H_0 : deaf children, hearing children 的 eye-movement rate 分布相同

H_a : 兩組分布不同

$$H_2 + \dots + 18 = \frac{18(19)}{2} = 171$$

$$F(W_D) = \frac{n_D(N+1)}{2} = \frac{9(19)}{2} = 85.5 < 126$$

$\therefore$ reject H_0

eyemovement dist differ between deaf / hearing

Two Simple Examples Use the sign test to compare two populations for significant differences for the paired data in Exercises 4–5. State the null and alternative hypotheses to be tested. Determine an appropriate rejection region with $\alpha \leq .10$. Calculate the observed value of the test statistic and present your conclusion.

4.

Population	Pair						
	1	2	3	4	5	6	7
1	8.9	8.1	9.3	7.7	10.4	8.3	7.4
2	8.8	7.4	9.0	7.8	9.9	8.1	6.9

+ + + - + + +

+6 -1

H_0 : two populations have no median difference

$$P(+)=P(-)=0.5$$

H_1 : medians are different

$$P(+)\neq 0.5$$

$\hat{X}$ = number of plus signs

$$X=6 \quad n=7$$

2-tailed p-value

$$P(X \geq 6 \text{ or } X \leq 1)$$

$$P(X \geq 6) = P(X=6) + P(X=7)$$

$$= \frac{C_6^7 + C_7^7}{2^7} = \frac{7+1}{128} = \frac{8}{128} = 0.0625$$

$$p = 2(0.0625) = 0.125 > 0.1 = \alpha$$

(fail to reject H_0)

there is not enough evidence to conclude the two populations differ.